School of Computer Science and Engineering
Vellore Institute of Technology
Vellore 632014, Tamil Nadu, India

October 31, 2026

The Editor-in-Chief
IEEE Transactions on Neural Networks and Learning Systems

Dear Editor-in-Chief,

We submit the manuscript *"Reliability-Aware Fusion of Token Evidence for Image-Text Verification in Unified Discrete Diffusion"* for consideration as a **Full Paper** in IEEE Transactions on Neural Networks and Learning Systems.

Unified masked diffusion models generate and understand text and images with one network, and with the image tokens masked and a caption visible, the same network assigns a probability to each observed image code. The manuscript studies image-text verification under image damage in one such pretrained model (UniDisc) on Common Objects in Context image-caption pairs. It learns a per-token reliability weighting of the model's own likelihood evidence, freezes it, and tests it once on fresh images in a preregistered, one-sided exact paired test under Gaussian noise at one level. The test confirmed the gain, including with length-matched wrong captions; on clean images the weighting stayed within a preregistered non-inferiority margin, which was fixed after an exploratory clean interval had been seen. The manuscript also reports, as exploratory, that a classical pixel denoiser applied before scoring and a linear logit on the same features reach comparable or higher accuracy in some conditions, so it does not claim that the learned additive form is the preferable defence. An exploratory sweep on the same images found the gain at milder and stronger Gaussian noise. The evidence covers one model, one dataset, and one tested noise level, and the paper states these limits.

The design images of the method come from an earlier preregistered study of the same model that tested a different question; its primary result was negative. That study is not part of this submission; the manuscript discloses it in its methods and deviations.

The study was preregistered in a dated analysis-plan addendum committed to a version-controlled repository before the test images were scored, with the frozen method pinned by hash, and every number in the paper is traced to committed result files. We believe the work fits the journal's scope in learning systems and information fusion, and that its preregistered design and its separation of confirmed from exploratory results will interest the readership.

The manuscript follows the journal's double-anonymous review policy: the review copy contains no author names, affiliations, funding or acknowledgment of the institution, and the code and data are available to reviewers in anonymized form. The supplementary material is provided as a separate file. We confirm that this work is original, has not been published elsewhere, and is not under review at any other publication. The authors declare no competing interests. All authors have approved the manuscript and agree with its submission.

Thank you for considering our manuscript.

Sincerely,

Subramaniyaswamy Vairavasundaram (corresponding author)
School of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, Tamil Nadu, India
E-mail: subramaniyaswamy.v@vit.ac.in ORCID: 0000-0001-5328-7672

on behalf of all authors:
Shlok Goenka (ORCID 0009-0002-8581-5657), Subramaniyaswamy Vairavasundaram (ORCID 0000-0001-5328-7672), and Balaji G N (ORCID 0000-0002-5346-0989)